In [6]:

```python
# 1. Import Libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.model_selection import train_test_split
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.naive_bayes import MultinomialNB
from sklearn.metrics import accuracy_score, classification_report
from sklearn.decomposition import PCA
from sklearn.cluster import KMeans
import string

# 2. Load Dataset
df = pd.read_csv(r"C:\Users\sonik\Desktop\Jupyter Files\iphone.csv")

# Quick look
print(df.head())
print(df.info())

# 3. Preprocess Text
# Using simple Python for cleaning
stop_words = set([
    'a','an','the','and','or','is','it','this','that','to','for','of','in','on','wi
    'as','its','at','from','by','be','are','was','were','but','i','my','me','you','
    'so','if','they','we','he','she','them','us'
])

def clean_text(text):
    if pd.isnull(text):
        return ""
    text = text.lower()  # lowercase
    text = text.translate(str.maketrans('', '', string.punctuation))  # remove punc
    text = ' '.join([word for word in text.split() if word not in stop_words])
    return text

# Combine title + description
df['review_text'] = df['reviewTitle'].fillna('') + ' ' + df['reviewDescription'].fi
df['review_text_clean'] = df['review_text'].apply(clean_text)

# 4. Sentiment Labeling
df = df[df['ratingScore'] != 3]  # ignore neutral
df['sentiment'] = df['ratingScore'].apply(lambda x: 'positive' if x > 3 else 'negat

# 5. Split Dataset
X = df['review_text_clean']
y = df['sentiment']

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_sta

# 6. Vectorize Text
tfidf = TfidfVectorizer(max_features=5000, stop_words='english')
X_train_tfidf = tfidf.fit_transform(X_train)
X_test_tfidf = tfidf.transform(X_test)
```

```python
# 7. Train Sentiment Classifier
model_nb = MultinomialNB()
model_nb.fit(X_train_tfidf, y_train)

# Predictions
y_pred = model_nb.predict(X_test_tfidf)

# Evaluation
print("Accuracy:", accuracy_score(y_test, y_pred))
print(classification_report(y_test, y_pred))

# 8. Feature Importance / Key Satisfaction Drivers
feature_names = tfidf.get_feature_names_out()
class_labels = model_nb.classes_
topn = 20

for i, class_label in enumerate(class_labels):
    top_features = np.argsort(model_nb.coef_[i])[::-1][:topn]
    print(f"\nTop words for {class_label} sentiment:")
    print([feature_names[j] for j in top_features])

# 9. Customer Segmentation
X_tfidf = tfidf.transform(df['review_text_clean'])
pca = PCA(n_components=2)
X_pca = pca.fit_transform(X_tfidf.toarray())

kmeans = KMeans(n_clusters=3, random_state=42)
df['cluster'] = kmeans.fit_predict(X_pca)

# Plot clusters
plt.figure(figsize=(8,6))
sns.scatterplot(x=X_pca[:,0], y=X_pca[:,1], hue=df['cluster'], palette='Set2')
plt.title("Customer Segments Based on Reviews")
plt.show()

# Top words per cluster
for c in df['cluster'].unique():
    text = ' '.join(df[df['cluster']==c]['review_text_clean'])
    words = text.split()
    top_words = pd.Series(words).value_counts().head(10)
    print(f"\nCluster {c} top words:")
    print(top_words)
```

```
  productAsin country        date  isVerified  ratingScore  \
0  B09G9BL5CP   India  11-08-2024        True            4
1  B09G9BL5CP   India  16-08-2024        True            5
2  B09G9BL5CP   India  14-05-2024        True            4
3  B09G9BL5CP   India  24-06-2024        True            5
4  B09G9BL5CP   India  18-05-2024        True            5

               reviewTitle                                 reviewDescription  \
0               No charger  Every thing is good about iPhones, there's not...
1          iPhone 13 256GB  It look so fabulous, I am android user switche...
2  Flip camera option nill  I tried to flip camera while recording but no ...
3                  Product                                        100% genuine
4             Good product         Happy to get the iPhone 13 in Amazon offer

                                            reviewUrl  \
0  https://www.amazon.in/gp/customer-reviews/R345...
1  https://www.amazon.in/gp/customer-reviews/R2HJ...
2  https://www.amazon.in/gp/customer-reviews/R3Y7...
3  https://www.amazon.in/gp/customer-reviews/R1P9...
4  https://www.amazon.in/gp/customer-reviews/R1XI...

                           reviewedIn                       variant  \
0  Reviewed in India on 11 August 2024  Colour: MidnightSize: 256 GB
1  Reviewed in India on 16 August 2024  Colour: MidnightSize: 256 GB
2     Reviewed in India on 14 May 2024  Colour: MidnightSize: 256 GB
3    Reviewed in India on 24 June 2024  Colour: MidnightSize: 256 GB
4     Reviewed in India on 18 May 2024  Colour: MidnightSize: 256 GB

  variantAsin
0  B09G9BQS98
1  B09G9BQS98
2  B09G9BQS98
3  B09G9BQS98
4  B09G9BQS98
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3062 entries, 0 to 3061
Data columns (total 11 columns):
 #   Column             Non-Null Count  Dtype
---  ------             --------------  -----
 0   productAsin        3062 non-null   object
 1   country            3062 non-null   object
 2   date               3062 non-null   object
 3   isVerified         3062 non-null   bool
 4   ratingScore        3062 non-null   int64
 5   reviewTitle        3062 non-null   object
 6   reviewDescription  2976 non-null   object
 7   reviewUrl          3046 non-null   object
 8   reviewedIn         3062 non-null   object
 9   variant            3062 non-null   object
 10  variantAsin        3062 non-null   object
dtypes: bool(1), int64(1), object(9)
memory usage: 242.3+ KB
None
Accuracy: 0.8424778761061947
              precision    recall  f1-score   support
```

```
    negative       0.98      0.40      0.57       146
    positive       0.83      1.00      0.90       419

    accuracy                           0.84       565
   macro avg       0.90      0.70      0.73       565
weighted avg       0.87      0.84      0.82       565
```

```
---------------------------------------------------------------------------
AttributeError                            Traceback (most recent call last)
Cell In[6], line 73
     70 topn = 20
     72 for i, class_label in enumerate(class_labels):
---> 73     top_features = np.argsort(model_nb.coef_[i])[::-1][:topn]
     74     print(f"\nTop words for {class_label} sentiment:")
     75     print([feature_names[j] for j in top_features])

AttributeError: 'MultinomialNB' object has no attribute 'coef_'
```

In [ ]: